

Research Assignment

SemEval-2027 Task 2: DiCo-NLI

Directional-Consistent Fine-Grained Natural Language Inference

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Setup

For this assignment, I evaluated two approaches to the English Track 1 DiCo-NLI task using the official training and development splits. The goal was to classify ordered text pairs into four possible relationships: **FORWARD_ENTAILMENT**, **BACKWARD_ENTAILMENT**, **EQUIVALENCE**, and **NEGATIVE_OTHER**. We evaluated all results on the official 660-item development set.

For System A, I initially attempted to fine-tune the recommended microsoft/deberta-v3-base model. However, training repeatedly crashed when I used FP16 mixed precision in Google Colab. DeBERTa-v3 was not stable with the FP16 setup I was using, which resulted in NaN losses and prevented the model from completing training. Running DeBERTa-v3 with full FP32 precision would have required more memory and training time than the available Colab resources could support.

Because the assignment allowed the use of distilbert-base-uncased when facing computational limitations, I switched to DistilBERT. The smaller model trained successfully with FP16 on the T4 GPU. I fine-tuned it for 4 epochs with a learning rate of 3×10^{-5} , batch size of 16, and maximum sequence length of 128. I also trained the model using two random seeds (42 and 123) to check whether its performance remained consistent across runs.

For System B, I took a different approach by evaluating Qwen2.5-1.5B-Instruct without fine-tuning. I compared zero-shot prompting with 6-shot prompting to examine whether providing examples improved the model's predictions and directional consistency. The 6-shot configuration was chosen specifically because the task requires classifying four distinct labels; providing 6 examples guaranteed full coverage of all four classes in the prompt while specifically demonstrating how directional labels flip under pair reversal (e.g., $A \rightarrow B$ as **FORWARD_ENTAILMENT** vs. $B \rightarrow A$ as **BACKWARD_ENTAILMENT**). Furthermore, 6 shots stayed strictly within the recommended 4–8 example range, balancing rich structural context with low inference latency and memory overhead on the T4 GPU. I also used a parser to convert the generated responses into the required labels; all outputs were successfully parsed.

Finally, I evaluated both systems using the three official task metrics: Weighted F1, SoftCons, and HardCons, allowing me to compare both classification performance and consistency when the order of a text pair was reversed.

Results

The results from the official development-set evaluation are summarized in Table 1. For System A, I report the individual results from Seeds 42 and 123, along with their mean and spread. For System B, I report the results from the zero-shot and 6-shot prompting configurations.

Table 1. Official Development-Set Results			
System	Weighted F1	SoftCons	HardCons
Pilot Reference (DeBERTa-v3-base)	0.7500	0.8400	0.7900
System A – DistilBERT (Seed 42)	0.5873	0.6137	0.5235
System A – DistilBERT (Seed 123)	0.6119	0.6426	0.5487
System A – Mean $\pm$ Spread	0.5996 ± 0.0123	0.6282 ± 0.0144	0.5361 ± 0.0126
System B – Qwen (Zero-Shot)	0.2088	0.0289	0.0144
System B – Qwen (6-Shot)	0.2410	0.3213	0.1264

Table 1 summarizes the official development-set results alongside the pilot DeBERTa-v3 baseline.

Analysis

System A clearly outperformed System B across all three evaluation metrics, as shown in Table 1. However, both systems fell short of the pilot DeBERTa-v3-base baseline (0.7500 Weighted F1, 0.8400 SoftCons, and 0.7900 HardCons). For System A, this performance gap may partly reflect differences in model capacity and architecture, as DistilBERT was substituted for DeBERTa due to hardware constraints. System B performed poorly overall, although 6-shot prompting substantially improved its directional consistency compared with zero-shot prompting, increasing SoftCons from 0.0289 to 0.3213 and HardCons from 0.0144 to 0.1264.

Figure 1 shows that the two systems exhibit distinct confusion patterns. System A performed best on BACKWARD_ENTAILMENT (73.7%), followed by FORWARD_ENTAILMENT (64.2%) and EQUIVALENCE (58.6%). Its weakest class was NEGATIVE_OTHER (39.6%), which was most often confused with FORWARD_ENTAILMENT (29.2%). In contrast, System B showed substantially greater difficulty with directional entailment and did not predict BACKWARD_ENTAILMENT for any example. Instead, 47.4% of FORWARD_ENTAILMENT and 46.8% of BACKWARD_ENTAILMENT examples were predicted as EQUIVALENCE. These results suggest that Qwen struggled to distinguish directional entailment relationships, whereas DistilBERT learned these distinctions more effectively but still had difficulty separating NEGATIVE_OTHER from FORWARD_ENTAILMENT.

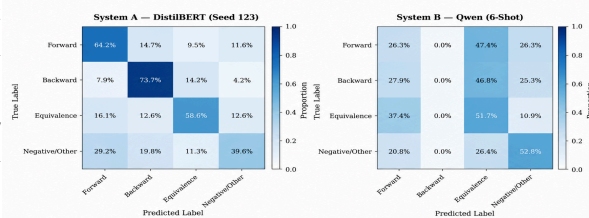


Figure 1. Normalized confusion matrices for System A (DistilBERT, Seed 123) and System B (Qwen, 6-Shot) on the development set.

To analyze the specific failure patterns of the stronger model, I examined 10 incorrect predictions from System A (Seed 123) (see Appendix A). A recurring issue was identifying the correct direction of entailment between closely related phrases. For example, “anti-protest laws” → “anti-protest law,” “Nine children” → “Five children,” and “Six killed” → “One dead” were labeled FORWARD_ENTAILMENT but predicted as BACKWARD_ENTAILMENT. High lexical and semantic similarity also appeared to cause confusion with EQUIVALENCE. For instance, both directions of “Pope Francis” and “pope,” as well as “in destroying” and “destroy,” were predicted as EQUIVALENCE despite having directional gold labels. The model also struggled with less direct semantic relationships, such as “of all Rage rallies” and “march of anger.” Overall, these errors suggest that DistilBERT generally recognized relationships between the phrases but often struggled to determine their precise direction.

These errors also raised the question of whether the model remained consistent when the same pairs were reversed. To examine this, I analyzed five cases where System A (Seed 123) failed SoftCons (see Appendix B). A clear pattern appeared in four of the five cases: DistilBERT correctly predicted BACKWARD_ENTAILMENT for the original pair but continued to predict BACKWARD_ENTAILMENT after reversal, even though the gold label changed to FORWARD_ENTAILMENT. This occurred for pairs such as “anti-protest law” → “anti-protest laws,” “Five children” → “Nine children,” “One dead” → “Six killed,” and “Brazil” → “Brazil crowds.” The remaining failure, involving “march of anger” and “of all Rage rallies,” changed from EQUIVALENCE to NEGATIVE_OTHER rather than following the expected directional transformation. These failures suggest that DistilBERT often recognized a directional relationship but did not reliably adjust its direction when the input order was reversed.

Taken together, these results show that System A was the stronger approach, but directional consistency remained an important limitation. DistilBERT generally distinguished the entailment classes more effectively than Qwen, yet it still struggled to adjust directional predictions reliably when closely related text pairs were reversed.

Next Steps

As a next step, I would test symmetric pair augmentation during training. For every training example (A, B), I would add the reversed pair (B, A) with the corresponding reversed label. For example, if (A, B) is labeled FORWARD_ENTAILMENT, then (B, A) would be labeled BACKWARD_ENTAILMENT. I expect this approach to improve directional consistency because the SoftCons analysis showed that DistilBERT often failed to adjust its predictions appropriately when the input order was reversed. Training explicitly in both directions could help the model learn the expected relationship between reversed pairs, potentially improving both SoftCons and HardCons. One limitation of this experiment was that the recommended DeBERTa-v3-base model could not be fully tested due to the available computational resources, so future work could also evaluate whether the same consistency issues occur with a stronger encoder.

Appendixes

Appendix A — 10 Error Cases

Error #1

Instance ID: dico_phrasis_0000004__en-en__flipped

Text 1: anti-protest laws

Text 2: anti-protest law

Gold Label: FORWARD_ENTAILMENT

Predicted Label: BACKWARD_ENTAILMENT

Error #2

Instance ID: dico_phrasis_0000010__en-en__flipped

Text 1: Nine children

Text 2: Five children

Gold Label: FORWARD_ENTAILMENT

Predicted Label: BACKWARD_ENTAILMENT

Error #3

Instance ID: dico_phrasis_0000018__en-en__flipped

Text 1: of all Rage rallies

Text 2: march of anger

Gold Label: FORWARD_ENTAILMENT

Predicted Label: NEGATIVE_OTHER

Error #4

Instance ID: dico_phrasis_0000018__en-en__original

Text 1: march of anger

Text 2: of all Rage rallies

Gold Label: BACKWARD_ENTAILMENT

Predicted Label: EQUIVALENCE

Error #5

Instance ID: dico_phrasis_0000021__en-en__flipped

Text 1: Six killed

Text 2: One dead

Gold Label: FORWARD_ENTAILMENT

Predicted Label: BACKWARD_ENTAILMENT

Error #6

Instance ID: dico_phrasis_0000023__en-en__flipped

Text 1: Pope Francis

Text 2: pope

Gold Label: FORWARD_ENTAILMENT

Predicted Label: EQUIVALENCE

Error #7
Instance ID: dico_phrasis_0000023__en-en__original
Text 1: pope
Text 2: Pope Francis
Gold Label: BACKWARD_ENTAILMENT
Predicted Label: EQUIVALENCE

Error #8
Instance ID: dico_phrasis_0000024__en-en__flipped
Text 1: in destroying
Text 2: destroy
Gold Label: FORWARD_ENTAILMENT
Predicted Label: EQUIVALENCE

Error #9
Instance ID: dico_phrasis_0000024__en-en__original
Text 1: destroy
Text 2: in destroying
Gold Label: BACKWARD_ENTAILMENT
Predicted Label: EQUIVALENCE

Error #10
Instance ID: dico_phrasis_0000031__en-en__flipped
Text 1: Brazil crowds
Text 2: Brazil
Gold Label: FORWARD_ENTAILMENT
Predicted Label: BACKWARD_ENTAILMENT

Appendix B — 5 SoftCons Failures

Failure #1
Text 1: anti-protest law
Text 2: anti-protest laws
Original: Gold = BACKWARD_ENTAILMENT | Predicted = BACKWARD_ENTAILMENT
Reversed: Gold = FORWARD_ENTAILMENT | Predicted = BACKWARD_ENTAILMENT

Failure #2
Text 1: Five children
Text 2: Nine children
Original: Gold = BACKWARD_ENTAILMENT | Predicted = BACKWARD_ENTAILMENT
Reversed: Gold = FORWARD_ENTAILMENT | Predicted = BACKWARD_ENTAILMENT

Failure #3
Text 1: march of anger
Text 2: of all Rage rallies
Original: Gold = BACKWARD_ENTAILMENT | Predicted = EQUIVALENCE
Reversed: Gold = FORWARD_ENTAILMENT | Predicted = NEGATIVE_OTHER

Failure #4
Text 1: One dead
Text 2: Six killed
Original: Gold = BACKWARD_ENTAILMENT | Predicted = BACKWARD_ENTAILMENT
Reversed: Gold = FORWARD_ENTAILMENT | Predicted = BACKWARD_ENTAILMENT

Failure #5
Text 1: Brazil
Text 2: Brazil crowds
Original: Gold = BACKWARD_ENTAILMENT | Predicted = BACKWARD_ENTAILMENT
Reversed: Gold = FORWARD_ENTAILMENT | Predicted = BACKWARD_ENTAILMENT
